FUNCTORIALITY OF THE DERIVED TENSOR PRODUCT AND ITS INVARIANCE UNDER QUASI-ISOMORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- The derived tensor product is computed by evaluating on K-flat complexes. discussed in the same section, source lines 14600-14726. Supplies the presentation of the functors used to define the transformation.
- Tensoring with a K-flat complex preserves quasi-isomorphisms. the author cites Lemma lemma-K-flat-quasi-isomorphism at proof lines 14749-14750. Gives that a quasi-isomorphism of the second argument induces an isomorphism of functors.

Source reference. Original lemma 0BYK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15018–15018. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 06Y0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8869–8869. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $R \to A$ be a ring map. Let $f : L^\bullet \to N^\bullet$ be a map of complexes of A-modules. Then f induces a transformation of functors*

$$1 \otimes f : - \otimes_A^{\mathbf{L}} L^\bullet \longrightarrow - \otimes_A^{\mathbf{L}} N^\bullet$$

If f is a quasi-isomorphism, then $1 \otimes f$ is an isomorphism of functors.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Since the functors are computing by evaluating on K-flat complexes $K^\bullet$ we can simply use the functoriality

$$\mathrm{Tot}(K^\bullet \otimes_R L^\bullet) \to \mathrm{Tot}(K^\bullet \otimes_R N^\bullet)$$

to define the transformation. The last statement follows from Lemma 06Y0. $\square$